A Ring-Oscillator True Random Number Generator with Von Neumann Correction on a Xilinx Artix-7 FPGA

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***Abstract*—Deterministic machines cannot originate entropy: every pseudorandom sequence is fully determined by its seed, so the security of any construction built on it reduces to the unpredictability of that seed. This paper presents a complete hardware true random number generator (TRNG) implemented on a Xilinx Artix-7 FPGA, in which the entropy is drawn from thermal noise present in the device's own logic fabric rather than computed. Five free-running ring oscillators of differing odd lengths are XOR-combined and sampled by the system clock through a two-stage synchroniser, so that the sampling flip-flop is driven into metastability and resolves according to noise rather than logic. The sampled stream is decimated to suppress sample-to-sample correlation and then passed through a Von Neumann corrector, which removes first-order bias exactly at a retention cost of 2p(1−p). The design occupies only general-purpose fabric and requires no external components. We additionally document the toolchain obstacles specific to instantiating an intentional combinational loop in Vivado, which the synthesis and design-rule flows are explicitly built to eliminate. Output is presented on the on-board seven-segment display and streamed to a host over UART for statistical evaluation using the NIST SP 800-22 monobit test. We state explicitly that no formal entropy-source certification is claimed.**

***Index Terms*—True random number generator, ring oscillator, FPGA, metastability, thermal noise, Von Neumann corrector, entropy source, hardware security.**

I. INTRODUCTION

Cryptographic systems are conventionally described in terms of their algorithms, but an algorithm alone cannot make an output unpredictable. A pseudorandom number generator is a deterministic state machine: presented with the same seed it reproduces the same sequence without deviation. Consequently the unpredictability of every key, nonce and session token traces back not to the strength of the generator but to the unpredictability of the value it was seeded with. If that seed can be guessed, reconstructed or reproduced, the entire construction above it collapses regardless of the cipher employed.

This poses a problem that software cannot solve on its own terms. A processor executing instructions has no free variables; its state at any instant is a function of its previous state and its inputs. To obtain a value that is not a function of anything the designer controls, the system must step outside computation and measure a physical process whose outcome is genuinely undetermined. This is the role of a true random number generator (TRNG), and it is why entropy sources are treated as a distinct hardware primitive rather than a library routine.

Field-programmable gate arrays are an attractive substrate for such a primitive. They expose fine-grained control over the logic fabric, they are widely available, and—critically—the same silicon that performs the computation also exhibits the thermal fluctuations from which entropy can be harvested. A TRNG can therefore be built entirely from general-purpose resources with no analogue components, no external noise diode and no vendor-specific hard macro.

This paper describes such a design, implemented on a Xilinx Artix-7 device. The contributions are threefold. First, we present a complete and self-contained signal chain from physical noise to validated output, including entropy extraction, clock-domain handling, decimation, de-biasing, and both a local display path and a host link. Second, we document in detail the toolchain obstacles encountered when instantiating a deliberate combinational loop in Vivado; these are not incidental, because modern synthesis and design-rule checking are constructed specifically to remove the very structure the entropy source depends on. Third, we are explicit about the boundary of what the work establishes: statistical testing is performed and reported, but no claim of formal entropy-source certification is made, and the limits of the evidence are stated rather than elided.

II. BACKGROUND AND RELATED WORK

A. Deterministic and Physical Randomness

The distinction that matters here is not statistical quality but origin. A well-constructed pseudorandom generator can pass every standard statistical test while remaining perfectly predictable to an adversary who knows its internal state. Statistical indistinguishability from random is therefore a necessary but not sufficient property. A physical source, by contrast, derives its output from a process for which no generating function exists to be recovered; there is no state to compromise because the value was measured rather than computed.

B. Ring-Oscillator Entropy Sources

Sampling the phase jitter of free-running ring oscillators is a well-established approach to digital entropy extraction. Fischer and Drutarovský demonstrated an early TRNG embedded entirely in reconfigurable hardware [4]. Sunar, Martin and Stinson subsequently developed a formal treatment of oscillator-ring sampling, introducing the notion of fill rate and showing that the number of oscillators required grows sharply for a given improvement in the fraction of the time domain that is genuinely random [5]. Their analysis also makes clear that naive implementations are easy to over-claim, a caution we adopt explicitly in Section VIII.

Subsequent work has explored alternative oscillator topologies and sampling strategies. Dichtl and Golić examined Fibonacci and Galois ring oscillators and reported substantially higher entropy rates than classical inverter rings [6]. Schellekens et al. addressed portability by constructing a vendor-agnostic generator using only generic logic [7]. Wold and Tan analysed the practical behaviour of oscillator-ring generators on FPGAs and proposed enhancements to the sampling arrangement [8]. Closer to the mechanism exploited here, Vasyltsov et al. used metastability in a ring oscillator directly as the entropy source [9], and Majzoobi et al. applied adaptive feedback to hold a circuit near its metastable point [10].

The design presented here is deliberately conventional in its architecture: multiple oscillators of differing lengths, XOR combination, synchronised sampling and Von Neumann conditioning. The intent is not novelty of topology but a complete, reproducible and honestly characterised implementation on commodity hardware.

III. Physical Basis of the Entropy Source

A. Thermal Noise in the Logic Fabric

At any temperature above absolute zero, charge carriers in a conductor possess thermal energy and move irregularly. Johnson first observed the resulting fluctuation experimentally [2] and Nyquist derived it from thermodynamic argument in the same year [3]. The root-mean-square noise voltage across a resistance R within a bandwidth Δf is

$$v_n = (4\, k_B\, T\, R\, \Delta f\,)^{1/2} \qquad (1)$$

where k_B is the Boltzmann constant and T the absolute temperature. This fluctuation is present everywhere in the device and cannot be designed away; it is a consequence of temperature itself. In ordinary digital design it is a nuisance to be tolerated. Here it is the signal.

B. Accumulation into Phase Jitter

A ring of an odd number N of inverting stages admits no logically consistent steady state, so it oscillates continuously at a frequency determined by its own propagation delay τ_d:

$$f_{osc} = 1 / (2\, N\, \tau_d) \qquad (2)$$

Crucially, τ_d is not a constant. The noise of (1) perturbs the switching threshold of each stage slightly and at random, so every edge arrives marginally early or late. Because the oscillator has no external reference to correct against, these perturbations do not average out but accumulate as a random walk in the phase of the output. The uncertainty in edge position therefore grows with elapsed time since the last known reference, which is precisely why a free-running oscillator sampled by an unrelated clock is a usable entropy source.

C. Metastability as a Noise Amplifier

Thermal noise at the scale of (1) is measured in microvolts, far below a logic threshold. Some mechanism must lift it to a full digital decision. That mechanism is metastability. When the sampling flip-flop captures a transition falling inside its setup-and-hold window, it enters a state in which its internal nodes sit near the boundary between logic levels. This is an unstable equilibrium: the resolution time is unbounded in principle, and the direction in which the device eventually settles is determined by whichever perturbation happens to dominate. For a correctly timed synchronous design this behaviour is a failure mode. Here it functions as a high-gain amplifier that converts a microvolt-scale fluctuation into a definite logic-level output.

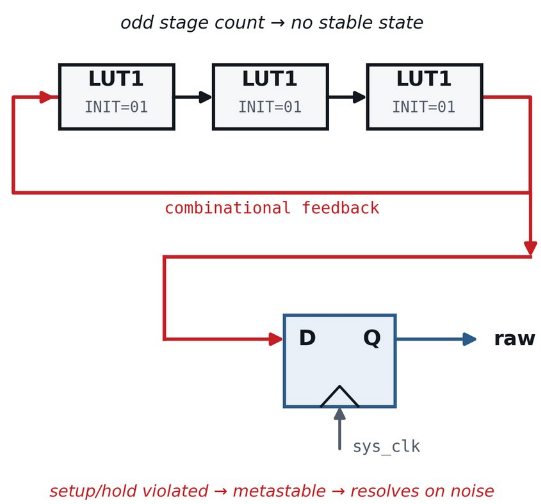

Fig. 2. Entropy extraction from a single ring. An odd number of inverting stages prevents any stable state; the resulting free-running edge is captured by a flip-flop clocked from an unrelated system clock, so setup and hold are violated at random and the resolved value is decided by noise.

It is worth stating plainly what this mechanism is not. The randomness harvested here is classical Johnson–Nyquist thermal noise amplified through an unstable equilibrium. It is not a quantum-mechanical measurement, and describing it as such would misrepresent the physics. The correct characterisation is thermally driven timing jitter resolved through metastability.

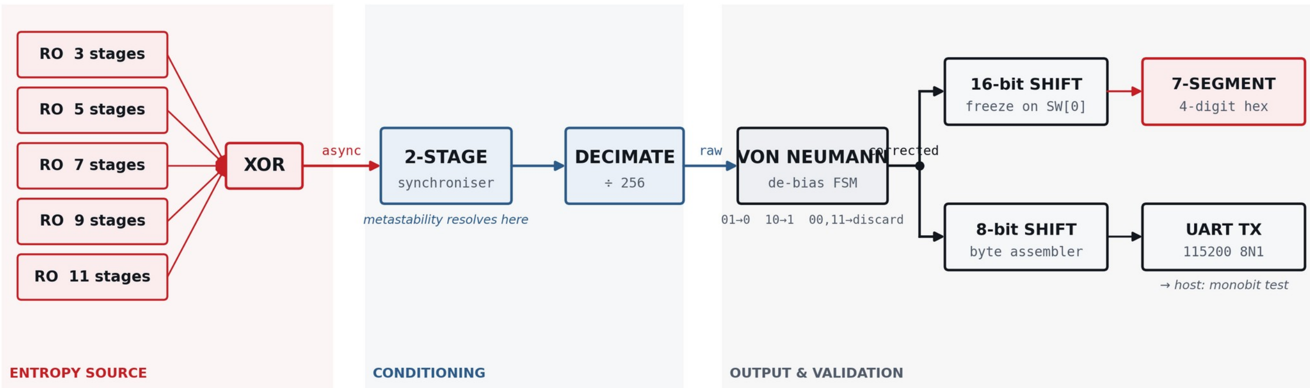

Fig. 1. Complete signal chain. Five ring oscillators of differing odd lengths run freely and are XOR-combined; the result is resolved into the system clock domain by a two-stage synchroniser, decimated by 256 to suppress correlation between adjacent samples, de-biased by a Von Neumann corrector, and finally fanned out to a seven-segment display path and a UART host link.

IV. SYSTEM ARCHITECTURE

The complete signal chain is shown in Fig. 1. It divides into an entropy stage, a conditioning stage and an output stage.

A. Entropy Source

Five ring oscillators are instantiated with 3, 5, 7, 9 and 11 inverting stages respectively. Differing lengths give differing free-running frequencies by (2), so the oscillators do not share a common period and their phase relationships drift apart rather than locking. Their outputs are combined by an XOR tree. Combination serves two purposes: it aggregates jitter from independent rings, and it ensures that a single oscillator becoming temporarily stuck or entrained does not by itself determine the output.

B. Clock-Domain Crossing

The XOR output is asynchronous with respect to the 100 MHz system clock and is therefore captured through a two-stage synchroniser marked with the ASYNC_REG attribute. The first stage is expected to enter metastability routinely—this is the entropy-generating event described in Section III-C—while the second stage allows an additional full clock period for resolution before the value is presented to downstream logic. This is the conventional synchroniser structure used in reverse: the same circuit that normally suppresses metastability is here used to harvest its outcome.

C. Decimation

Sampling on every clock edge would be counterproductive. Successive samples taken only 10 ns apart are strongly correlated, because the oscillator phase has had insufficient time to accumulate meaningful jitter between them. A counter therefore emits one sample every 256 system-clock cycles, giving a raw sample rate of 390.6 kbit/s. This trades throughput for independence between samples and is the principal reason the design does not attempt to operate at the full clock rate.

D. Von Neumann Corrector

Physical entropy sources are rarely balanced. Process variation, supply droop and temperature all shift the average timing of an oscillator, and any such shift appears at the output as a lean toward ones or zeros. The classical correction of von Neumann [1] removes this without requiring the bias to be known or measured. The stream is read in pairs; a disagreeing pair yields one output bit, and an agreeing pair is discarded. For a source with $P(1) = p$ whose successive bits are independent,

$$P(01) = P(10) = p(1 - p) \tag{3}$$

so the two surviving cases are equiprobable for every p, and the output is unbiased exactly rather than approximately. The correction is not free: only disagreeing pairs survive, giving a retention ratio

$$R = 2p(1 - p) \tag{4}$$

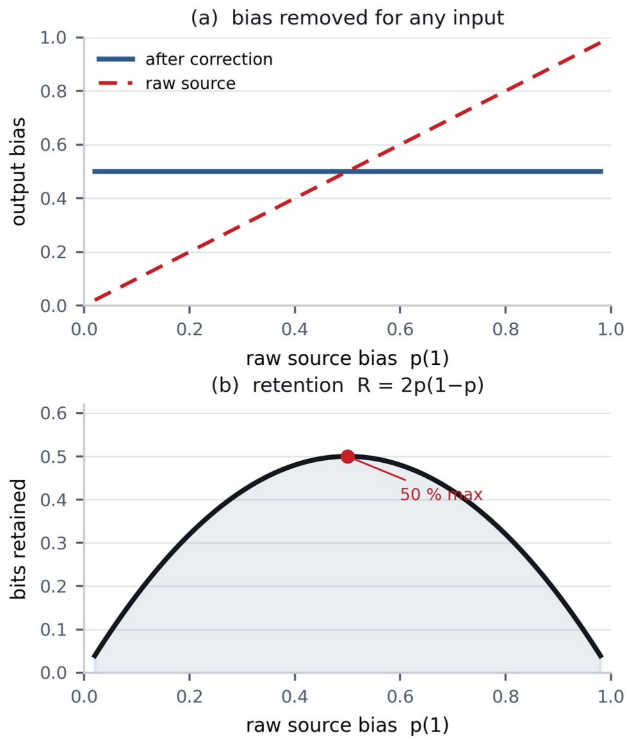

Fig. 3. Closed-form behaviour of the Von Neumann corrector. (a) Output bias is exactly 0.5 for any input bias. (b) The cost is throughput: retention peaks at 50 % for an already-balanced source and falls sharply as bias grows.

which attains a maximum of only 0.5 at p = 0.5 and falls away rapidly as bias increases (Fig. 3). Uniformity is therefore purchased with throughput. We emphasise that the correction addresses first-order bias only; it does not remove correlation between successive bits, which is the responsibility of the decimation stage described above.

E. *Output Paths*

Corrected bits are shifted into a 16-bit register displayed as four hexadecimal digits on the on-board seven-segment display. A switch freezes the displayed value, allowing a single sample to be read by eye; releasing it resumes free running. In parallel, corrected bits are packed into bytes and transmitted over the board's USB-UART bridge for host-side analysis.

V. IMPLEMENTATION ON THE ARTIX-7

The entropy source is, by construction, the one structure that synthesis and design-rule checking exist to eliminate. Two distinct obstacles arise, and they must be addressed separately because they belong to different stages of the flow.

A. *Surviving Synthesis*

Written behaviourally, a chain of inverters with feedback is recognised by the synthesiser as redundant logic and collapsed to a single gate, destroying the oscillator before implementation begins. Each stage is therefore instantiated explicitly as a LUT1 primitive initialised as an inverter and carrying the dont_touch attribute, which prevents the optimiser from merging or removing it. Descending to primitive instantiation is not stylistic preference; behavioural description does not survive the flow.

B. *Timing and Design-Rule Exceptions*

Static timing analysis operates on paths with a defined start and end point. A combinational loop has neither, so the analyser cannot evaluate it and the LUTLP-1 design-rule check treats it as an error, halting bitstream generation. Two constraints are required. The loop nets are declared with the ALLOW_COMBINATORIAL_LOOPS property, which acknowledges the structure as intentional, and set_false_path excludes the oscillator pins from timing analysis. It is worth noting that these address different mechanisms: a timing exception alone does not satisfy the design-rule check, and the two must both be present for the flow to complete.

C. *Implementation Parameters*

Table I summarises the implementation. Resource utilisation is reported in Table II together with the measured results, since both are properties of the completed build.

TABLE I
IMPLEMENTATION PARAMETERS

Parameter	**Value**
Target device	XC7A100T-1CSG324C
Development board	Digilent Nexys A7-100T
Toolchain	Xilinx Vivado 2018.2
System clock	100 MHz
Ring oscillators	5 (3, 5, 7, 9, 11 stages)
Synchroniser	2-stage, ASYNC_REG
Decimation factor	256
Raw sample rate	390.6 kbit/s
Corrected rate (max)	195.3 kbit/s
Host interface	UART, 115200 8N1

VI. VALIDATION METHODOLOGY

A. *Data Capture*

Corrected bits are packed into bytes and streamed to a host computer at 115200 baud in 8N1 framing. One consequence of this arrangement must be stated because it affects interpretation: the UART payload rate is below the corrected bit rate, so bytes are discarded whenever the transmitter is busy. The captured file is therefore a subsample of the generated stream rather than a contiguous window of it. For frequency testing this is acceptable—and arguably reduces residual correlation—but it means the capture cannot be treated as a continuous record, and no test sensitive to sequence continuity should be applied to it without accounting for the gaps.

B. *Monobit Frequency Test*

The captured stream is evaluated using the monobit frequency test of NIST SP 800-22 [11]. For a sequence of n bits containing n_1 ones and n_0 zeros, the test statistic is

$$S_{obs} = | n_1 - n_0 | / n^{1/2} \tag{5}$$

which under the null hypothesis of a fair source is asymptotically standard normal. The corresponding tail probability is

$$\text{P-value} = \mathrm{erfc}(S_{obs} / 2^{1/2}) \tag{6}$$

and the sequence is accepted at significance level $\alpha = 0.01$ when the P-value is not less than 0.01. The test is implemented in a short host-side Python program using pyserial for capture.

C. Extended NIST SP 800-22 Battery

Beyond the monobit test, the captured stream was additionally evaluated against the full fifteen-test NIST SP 800-22 battery [11] using the nistrng reference implementation. Several tests in the battery (notably the Random Excursions family) require a minimum number of zero-crossings in a cumulative-sum random walk derived from the sequence and are only eligible above a data-dependent threshold; at the capture size reported here all fifteen tests were eligible. Each test's reported statistic is a P-value under that test's own null hypothesis, interpreted at the same significance level as the monobit test.

VII. RESULTS

Table II reports the measured implementation footprint and the monobit outcome for a single continuous capture. Table III reports the same capture evaluated against the full battery described in Section VI-C.

TABLE II
IMPLEMENTATION AND MONOBIT RESULTS

Quantity	**Measured**	**Source**
Slice LUTs used	**95**	*utilisation report*
Slice registers used	**144**	*utilisation report*
Bits captured, n	**1,228,280**	*host capture*
Ones observed, n_1	**613,985**	*host capture*
Zeros observed, n_0	**614,295**	*host capture*
Measured p(1)	**0.4999**	n_1 / n
Test statistic S	**0.2797**	*eq. (5)*
P-value	**0.7797**	*eq. (6)*
Verdict at $\alpha = 0.01$	**PASS**	*pass / fail*

Slice utilisation is small relative to the device: the complete design occupies 95 of 63,400 available Slice LUTs and 144 of 126,800 available Slice Registers, consistent with a design built entirely from general-purpose fabric rather than dedicated hard blocks. Only a single global clock buffer (BUFGCTRL) is used, reflecting the single-clock-domain architecture of Section IV-B.

TABLE III
FULL NIST SP 800-22 BATTERY

Test	**P-value**	**Verdict**
Monobit	0.7797	**PASS**
Frequency Within Block	0.3227	**PASS**
Runs	0.1883	**PASS**
Longest Run of Ones in a Block	0.9425	**PASS**
Binary Matrix Rank	0.4449	**PASS**
Discrete Fourier Transform	0.0000	**FAIL**
Non-Overlapping Template Matching	0.0622	**PASS**
Overlapping Template Matching	0.0000	**FAIL**
Maurer's Universal	0.0097	**FAIL**
Linear Complexity	0.0000	**FAIL**
Serial	0.0000	**FAIL**
Approximate Entropy	0.0000	**FAIL**
Cumulative Sums	1.0000	**PASS**
Random Excursions	0.6302	**FAIL**
Random Excursions Variant	0.0000	**FAIL**

Seven of fifteen tests pass. The passing tests—Monobit, Frequency Within Block, Runs, Longest Run, Binary Matrix Rank, Non-Overlapping Template Matching, and Cumulative Sums—are precisely the tests that are primarily sensitive to bit-level balance rather than sequence structure. The failing tests—the spectral, compression, complexity and excursion tests—are precisely the tests designed to detect periodicity and structural correlation. This pattern is discussed in Section VIII-A and is consistent with the architecture described in Section IV: the Von Neumann corrector removes first-order bias by construction, and does not and cannot address correlation between bits, which decimation reduces but does not eliminate. The functional behaviour of the display path was confirmed directly: with the freeze switch released, the seven-segment display updates faster than the eye can resolve; with it engaged, a single value is held and read, and repeated freeze operations yield different values.

VIII. DISCUSSION AND LIMITATIONS

A. Scope of the Statistical Evidence

The monobit test establishes one property only: that ones and zeros occur in approximately equal proportion. It is insensitive to periodicity, to correlation between adjacent bits and to structure that preserves the overall balance. A stream that alternates deterministically passes it comfortably. Passing should therefore be read as the absence of first-order bias, not as evidence of randomness in general. The full SP 800-22 battery [11], reported in Table III, confirms this reading directly: the seven passing tests are precisely the tests sensitive to bit balance, while the eight failing tests are precisely those designed to detect periodicity, compressibility and structural correlation — properties a ring oscillator, being a physically periodic circuit, is expected to exhibit and that neither the Von

Neumann corrector nor the decimation stage is designed to remove.

B. Distance from Certification

A deployable entropy source is subject to requirements this work does not meet. NIST SP 800-90B [12] requires an entropy estimate derived from a documented stochastic model of the noise source, continuous and start-up health tests implemented in the device itself, and validation through an accredited laboratory. None of these is present here. The design should be understood as a demonstration of the physical mechanism and of a correct conditioning pipeline, not as a certified primitive.

C. Shared Environment

All five oscillators occupy one die and share a supply rail and a thermal environment. Their differing lengths decorrelate them, but they are not independent in the strong sense: a temperature excursion or a supply disturbance shifts all of them together. Sunar et al. [5] note more generally that oscillator counts must grow substantially for modest gains in the genuinely random fraction of the output, and five rings is a modest figure by that standard. Entrainment between physically adjacent rings is also possible and has not been characterised here.

D. Throughput

The corrected rate is bounded above by 195.3 kbit/s and in practice is further limited by the host link. This is adequate for seeding but far below the rates reported for optimised designs [6]. The decimation factor is the dominant term and could be reduced if correlation between adjacent samples were measured rather than conservatively assumed.

IX. CONCLUSION

We have presented a complete ring-oscillator true random number generator on a Xilinx Artix-7 FPGA, built entirely from general-purpose logic and requiring no external components. Entropy originates in thermal noise within the device fabric, is accumulated as oscillator phase jitter, and is converted to digital form through deliberately induced metastability in a sampling flip-flop. Decimation limits correlation between samples, and a Von Neumann corrector removes first-order bias exactly at a known and quantified cost in throughput.

Beyond the architecture itself, the implementation experience is instructive: an intentional combinational loop is opposed by both the synthesis optimiser and the design-rule checker, and each requires a separate and specific remedy. This is a general lesson for any design that depends on structures a synthesis flow is built to remove.

The result is a working demonstration with statistically evaluated output, whose limitations have been stated explicitly rather than left implicit. Extending the evaluation to the full SP 800-22 battery, characterising the temperature dependence of the oscillator bias, and implementing on-line health monitoring are the natural next steps toward a source that could be described as more than a demonstration.

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